```
  0 1 2 3 4
0 * * * * *
1   * * * *
2     * * *
3       * *
4         *
```

```
int n=5
for(int i=0;i<n;i++){
        for (int j=0;j<i;j++){
            so.p(" ");
        }
        For (int j= n-1; j>=i; j--){
            so.p("*");
        }
        soP();
}
```

n=5

i=0 — 0 < 5 — i++ [1]

j=0 — 0 < 0 — ✗

j=n-1 — j>=i — j--

j=4 — 5>=0 — * [3]

j=3 — 3>=0 — * [2]

j = 2 2 >= 0 [2]

*

j = 1 1 >= 0 [0]

*

j = 0 0 >= 0 -1

*

j = -1 -1 >= 0

*

i = 1 1 < 5

j = 0 j < i [1]

0 < 1

" "

j = 1 1 < 1

*

j = n - 1 j >= i j--

j = 4 4 >= 1 [3]

*

j = 3 3 >= 1 [2]

*

j = 2 2 >= 1 [1]

*

j=1 1>=1 ✓ 0

j=0 0>=1 ✗

جـ

i=2 2<5

j=0 j<1 j++ 1

0<2 ✓

j=1 1<2 ✓ 2

j=2 2<2 ✗

j=n-1 j>=1 j--

j=4 4>=2 ✓ 3

j=3 3>=2 ✓ 2

j=2 2>=2 ✓ 1

j=1 1>=2 ✗

```
  0 1 2 3 4
0 1 2 3 4 5
1 1 2 3 4 5
2 1 2 3 4 5
3 1 2 3 4 5
4 1 2 3 4 5
```

```
int n=5
for (int i= 1 ; i<=n ; i++ ) {
        for (int j=1 ; j<=n ; j++) {
                S.O.P(j);
        }
        S.O.P();
}
```

int i=1   1<=5   i++   2

j=1   j<=5   2

1<=5

1

j=2   2<=5   3

2

j=3   3<=5   4

3

j=4   4<=5   5

4

1 2 3 4 5

1 2 3 4 5

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j=5 5<=5 ⑥

5

j=6 6<=5

x

i=2 2<=5 i++ ③

j=1 1<=5 ②

1

j=2 2<=5 ③

2

j=3 3<=5 ④

3

j=4 4<=5 ⑤

4

j=5 5<=5 ⑥

5

j=6 6<=5

x

i=3 3<=5 (4)

j=1 1<=5 (2)

1

j=2 2<=5 (2)

2

j=3 3<=5 (2)

3

j=4 4<=5 (2)

4

j=5 5<=5 (2)

5

j=6 6<=5

x

3)

	0	1	2	3	4
0	*	*	*	*	*
1	*	*	*	*	*
2	*	*	*	*	*
3	*	*	*	*	*
4	*	*	*	*	*

```
int n=5;
for(int i=0; i<n; i++){
    for(int j=0; j<n; j++){
        sop("*");
    }
    ~~sopln~~
}
```

int i=0 i<n i++ ①

0<5

j=0 j<5 ①

0<5

j=1 1<5 ②

j=2 2<5 ③

j=3 3<5 ④

j=4 4<5 ⑤

j=5 5<5 ✗

i=1 1<5

j=0 0<5 ①

j=1 1<5 ②

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j=2 2<5 ①

j=3 3<5 ④

j=4 4<5 ⑤

j=5 5<5 ✗

i=2 2<5 i++ ③

j=0 0<5 ①

j=1 1<5 ②

j=2 2<5 ③

j=3 3<5 ④

j=4 4<5 ⑤

j=5 5<5 ✗

$i = 3$ $\quad 3 < 5$ $\quad i++$ ④

$j = 0$ $\quad 0 < 5$ ②

$j = 1$ $\quad 1 < 5$

$j = 2$ $\quad 2 < 5$

$j = 3$ $\quad 3 < 5$

$j = 4$ $\quad 4 < 5$

$j = 5$ $\quad 5 < 5$

$i = 4$ $\quad 4 < 5$

$j = 0$ $\quad 0 < 5$

$1 < 5$

$2 < 5$

$i = 5$ $\quad 5 < 5$